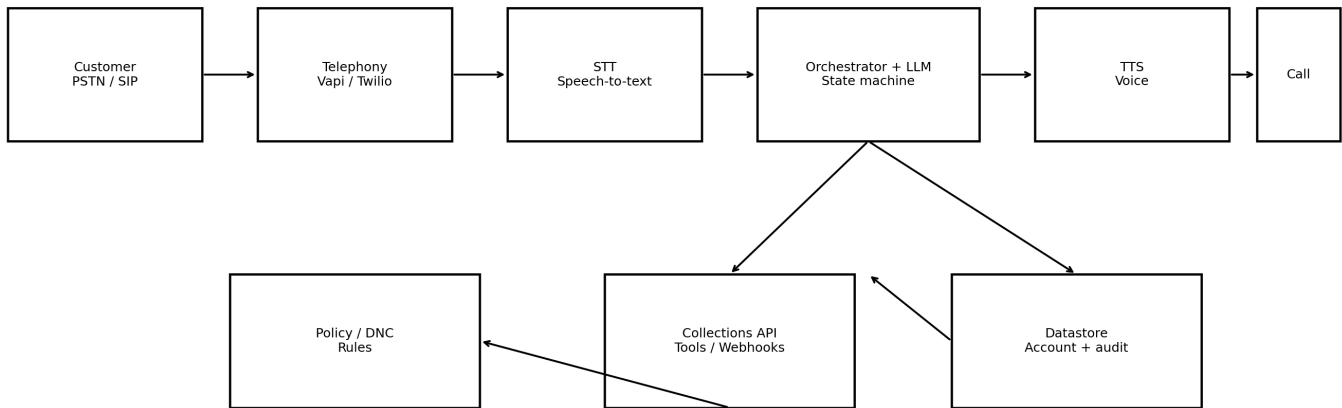


Kapture Finance Collections Voicebot — High-Level Design

Engineer handoff design for an outbound overdue-EMI voice agent. Example context: Rahul Sharma, personal loan, overdue EMI ₹8,499, 12 DPD.

Kapture Finance Collections Voicebot — logical architecture



Target latency: STT 150–300 ms • Orchestrator/LLM 300–700 ms • Tool API 100–300 ms • TTS first audio 200–500 ms • turn-to-first-audio target <1.5 s

1. Architecture & latency budget

Pipeline: telephony → Vapi media/turn handling → streaming STT → deterministic collections state machine + LLM response planner → tool/API layer → TTS → telephony. The state machine owns authorization to disclose debt; the LLM cannot override it. Suggested per-turn budget: STT 150–300 ms; orchestration/LLM 300–700 ms; tool/API 100–300 ms; TTS first audio 200–500 ms; target end-to-first-audio under 1.5 s. Keep prompts short, stream STT/TTS, cache policy/configuration, and make tool calls idempotent.

2. State machine & locks

INIT → DISCLOSURE → AUTH_PENDING → AUTHENTICATED → PURPOSE_AND_DUE → INTENT → NEGOTIATION/RESOLUTION → CLOSING → DISPOSITION. AUTH_PENDING is locked: no balance, DPD, loan type, payment status, or collection reason may be disclosed until verify_customer returns verified=true. WRONG_PERSON, DNC, ALREADY_PAID, DISPUTE, HARDSHIP, HOSTILE, CALLBACK, VOICEMAIL and SYSTEM_ERROR are terminal/branch states with explicit dispositions. A policy engine, not the prompt alone, enforces disclosure and DNC rules.

3. Intents & entities

Primary intents: will_pay, cannot_pay_hardship, dispute, already_paid, wrong_person, callback_request, hostile, do_not_call, payment_link_request, partial_pay, needs_human, no_input/voicemail. Extract only necessary entities: promised amount, PTP date/time, preferred language, dispute reason/category, callback date/time, payment channel, and authentication result. Never infer a PTP date or amount.

4. Tools / APIs

get_account_details(account_ref) → masked account state and permitted fields. verify_customer(customer_ref, verification_type, verification_value) → verified boolean plus verification method. log_promise_to_pay(customer_ref, amount, due_date) → PTP id/status. send_payment_link(customer_ref, amount, channel) → link/status. escalate_to_agent(customer_ref, reason, priority) → transfer/ticket status. mark_disposition(call_id, outcome, reason, next_action_at) → audit status. Tools reject calls inconsistent with state, especially any debt-revealing response before authentication.

5. Authentication & data safety

Use a non-sensitive lookup such as the dialed-number/customer reference to retrieve a candidate record, then verify with an approved factor such as date of birth fragment, PIN/OTP, or another policy-approved knowledge factor. If someone else answers, say only that Maya is calling from Kapture Finance and ask whether the intended customer is available; do not mention loan, EMI, overdue, balance, or collections. Mask PII in logs, encrypt data in transit/at rest, minimize retention, apply role-based access, and use authenticated Vapi server URLs.

6. Guardrails & fair collections

Mandatory opening identifies Maya, Kapture Finance, and the purpose at the permitted disclosure stage. Follow applicable calling windows and consent/opt-out rules for the deployment jurisdiction. No threats, shame, coercion, deception, repeated harassment, or fabricated consequences. Never invent fees, penalties, legal action, credit impact, settlement terms, or deadlines. Respect DNC immediately. If the customer disputes the debt/amount, stop negotiation and route to the dispute workflow/human. Off-topic requests are redirected once, then ended or transferred.

7. Edge cases

Already paid → acknowledge, avoid arguing, record already_paid and request proof/reference only if policy permits. Dispute → record dispute category and escalate; do not pressure payment. DNC → apologize, log DNC, stop outbound collection activity. Wrong number/person → no debt disclosure, mark wrong_person after safe confirmation. Voicemail/no input → leave only an approved neutral callback message if permitted; otherwise disposition no_contact. Hostile caller → remain calm, offer human escalation or end call; never retaliate. EN/HI switch → confirm preferred language and continue only with supported approved script; do not translate policy terms ad hoc.

8. Escalation & disposition

Human handoff for disputes, verified hardship requiring assistance, settlement requests outside configured authority, repeated authentication failure, legal/complaint threats, accessibility/language limitations, or explicit human request. Every call ends with mark_disposition, with outcome, reason, authentication state, PTP fields if any, DNC flag, transfer status, and next action timestamp. Do not store raw sensitive verification values.

9. Observability

Log call id, timestamps, state transitions, tool names/results, latency, disposition, language, transfer reason, authentication success/failure, PTP amount/date, and policy blocks. Redact sensitive values. Metrics: answer rate, containment, PTP rate, PTP kept rate, payment-link conversion, dispute rate, DNC rate, human-transfer rate, wrong-number rate, average turn latency, p95 latency, call drop rate, authentication failure rate, tool error rate, and hallucination/policy-violation rate.

10. Vapi implementation notes

Create an Assistant, attach a phone number, and add custom function tools with public server URLs. Vapi supports custom function tools and server URLs; function calls are delivered as tool-calls events and the server responds with results keyed by toolCallId. For local development, expose the webhook with a tunnel and optionally use the Vapi CLI forwarder. Use a low-temperature model, streaming transcription, a natural but professional voice, and short spoken turns. The recommended build separates the deterministic policy/state layer from the LLM so prompt drift cannot unlock debt disclosure.

Appendix — Example call flow

Unverified: “Hi, I’m Maya calling from Kapture Finance. May I speak with Rahul Sharma?”

Verify: customer provides approved verification factor → tool confirms verified.

Disclose: “Thank you for verifying. I’m calling about an overdue personal-loan EMI of ■8,499.”

Intent: ask whether they can pay now, by a specific date, or need help.

PTP: confirm exact amount and date, call log_promise_to_pay, then send payment link if requested.

Dispute: stop collection negotiation, log/escalate dispute, close politely.

Close: state the agreed next step and disposition the call.